

Assignment: Identifying an Unmet Clinical Need and Proposing a Robotic Solution

Medical Robotics – Prof. Marilena Vendittelli

1. Overview

You will identify a real clinical problem or limitation, analyze why it remains unsolved, and propose a robotic solution at the concept level. The goal is not to invent a fully engineered system. The goal is to demonstrate that you can think rigorously about where robotics genuinely adds value in medicine and where it does not.

A good submission could plausibly become the first slide deck of a startup pitch, a grant proposal aim, or a thesis introduction. A weak submission describes a vague problem and proposes a vague robot.

2. Deliverable

A written report of max 6 pages including figures and references, structured according to the sections below. Submit as a PDF file.

3. Required Sections

1. Clinical need (quantified)

Define the clinical problem in one or two sentences that a non-clinician can understand, then quantify it. Provide at least two of:

- Annual procedure volume or disease incidence.
- Complication or failure rate.
- Mortality or morbidity attributable to the problem.
- Direct cost, length of stay, or workforce gap.

Use peer-reviewed sources, national registries, [World Health Organization \(WHO\)](#) and [CDC](#) data, or other qualified sources. At least three independent citations are required for this section.

A clinical need is not “patients suffer from X.” A clinical need is “X affects N patients per year in [population], current standard of care fails in $M\%$ of cases, with consequences A, B, and C.”

2. Current standard of care and its limitations

Describe how the problem is managed today. Identify the specific failure mode you intend to address. If a robotic system already exists, describe it and explain what it does *not* do — clinically, technically, or economically — and why that gap matters. Vague claims that an existing robot is “too expensive” or “not accurate enough” are insufficient. Be specific.

3. The human limitation being addressed

Name the human limitation your robot relieves. Examples include sub-millimeter precision, tremor, radiation exposure, fatigue across long procedures, force control below human haptic threshold, presence at bedside, consistency across operators, availability of trained personnel, or working in spaces too small for human hands. If you cannot name a specific limitation, the problem may not need a robot. Reconsider before proceeding.

4. Proposed robotic solution

Describe your concept at a level appropriate to a Master's-level engineering audience. Cover, at minimum:

- Kinematic architecture and where it sits in the clinical workflow (bedside, OR-integrated, mobile, wearable, implantable, etc.).
- Degrees of freedom and the rationale for them.
- Sensing required (imaging modality, force, position, physiologic) and why.
- Level of autonomy (0 = none, 5 = full autonomy, see slides on classification). Justify your choice.
- Human-in-the-loop interface: who supervises, where, and how they intervene.
- One labeled sketch or diagram of the concept.

Avoid simply naming an existing platform with a new application stuck on it. If your concept is “a da Vinci that also does X,” you need to show why this is the best solution.

5. Why this is feasible now

Identify the recent enabling technologies — imaging, actuation, materials, AI/perception, miniaturization, regulatory pathway — that make this solution possible today, when it was not five or ten years ago. If you cannot identify any, the idea may be premature or may not need a robot at all.

6. Regulatory and adoption pathway

One paragraph each:

- Likely regulatory class (FDA Class II/III or EU MDR equivalent) and closest predicate device, if any.
- Identify the payer and whether your device replaces an existing line item or creates a new one. “It will save money for the system” is not a payment model.

7. Risks and failure modes

List the three most likely reasons your solution would fail — clinical, technical, or commercial — and how you would mitigate each. Honest analysis here is rewarded.

8. References

About 10 peer-reviewed sources. Vendor websites, press releases, and Wikipedia do not count toward the minimum, though they may be cited as supplementary.

4. Scope Guidance

Pick a problem you can describe end-to-end in the page limit. A narrowly defined problem (“bile duct injury during laparoscopic cholecystectomy in trainee-led cases”) almost always produces a stronger paper than a broad one (“surgical errors”). You may choose any clinical area.

Problem types that work well:

- A high-volume routine procedure where outcome variability is the issue.
- A low-volume high-stakes procedure where skill is concentrated and access is limited.
- A procedure currently done blind or by feel where imaging exists but is not integrated.
- A procedure where operator harm (radiation, ergonomics, infection) is a recognized problem.

Problem types that work less well:

- Anything where the bottleneck is a drug, a diagnostic test, or a policy decision rather than a physical manipulation.
- Problems already well-served by existing robots where your only contribution is a software tweak.
- Problems so vast that no single device could address them (“aging”).

5. What I Am Looking For

A clear chain of reasoning from a specific clinical problem, through a named human limitation, to a robotic solution whose form follows from that limitation. **The strongest reports will read as if the solution were inevitable given the problem. The weakest will read as if the solution were chosen first.**

I will reward intellectual honesty about what is hard, what is uncertain, and what your concept does not solve. I will penalize hand-waving, especially around regulatory pathway, and the question of whether a robot is actually the right answer.

6. Logistics

- Submit as a single PDF named `Surname_Firstname_ClinicalNeed.pdf`.
- Due: *[31/05/2026]*. You will present it at the exam date, after the group project presentation.
- Individual work.
- AI tools use: permitted but **must** be disclosed.

A. Catalogue of Candidate Clinical Needs

The following is a non-exhaustive catalogue of clinical problems where robotic support may be valuable. **The catalogue has mostly been generated using the LLM Claude and not all the numbers have been verified.**

You are not required to choose from this list, but it indicates the level of specificity expected. Items marked **(no robotic solution exists)** are open problems, **but this does not mean that a robotic solution would be appropriate**; your task is to find out if this is the case and then, possibly, propose a solution. Items marked **(partial robotic solution)** have commercial systems available, and your task in those cases is to identify the specific gap the existing system does not address.

Surgical Procedures

High-volume general and abdominal surgery.

- Cesarean section — approximately one in three deliveries globally; hemorrhage, bladder injury, and uterine closure quality are operator-dependent. **(no robotic solution exists)**
- Laparoscopic cholecystectomy — bile duct injury occurs in 0.3–0.7% of cases despite mature laparoscopic technique. **(partial robotic solution)**
- Appendectomy in standard cases — among the most common abdominal operations worldwide; existing robots are economically unjustified. **(partial robotic solution)**
- Inguinal and ventral hernia repair — mesh placement, fixation, and recurrence prevention remain technique-dependent. **(partial robotic solution)**
- Vaginal hysterectomy and pelvic floor repair — guideline-recommended when feasible but technically demanding in narrow working space. **(no robotic solution exists)**
- Routine surgical closure and suturing — ubiquitous, time-consuming, quality varies. **(no robotic solution exists)**

Specialty surgery.

- Pediatric and neonatal minimally invasive surgery — existing platforms are not scaled for centimeter-scale working spaces. **(no robotic solution exists)**
- Microsurgery in confined corridors — cochlear implantation, retinal vein cannulation, super-microsurgical lymphaticovenous anastomosis. **(no robotic solution exists)**
- Hand and digit replantation, nerve repair — time-critical, fatigue-sensitive, sub-millimeter anastomoses. **(no robotic solution exists)**
- Tonsillectomy and adenoidectomy — the most common pediatric operation; post-tonsillectomy hemorrhage is significant. **(no robotic solution exists)**
- Cataract surgery — approximately 30 million cases annually; phacoemulsification and IOL placement remain manual. **(partial robotic solution)**
- Skin cancer excision and Mohs surgery — enormous and growing volume; serial margin assessment is slow. **(no robotic solution exists)**
- Lumpectomy and sentinel lymph node biopsy — intraoperative localization of non-palpable lesions and margin assessment are the rate-limiting steps. **(no robotic solution exists)**
- Thyroidectomy and parathyroidectomy — recurrent laryngeal nerve injury and hypoparathyroidism remain feared complications. **(no robotic solution exists)**
- Carpal tunnel release, trigger finger, and common hand procedures — high volume, office-based. **(no robotic solution exists)**
- Total knee and hip arthroplasty — existing platforms address bony cuts but not soft tissue balancing. **(partial robotic solution)**
- Spine fusion beyond pedicle screw placement — decompression, interbody preparation, deformity correction. **(partial robotic solution)**
- Stereotactic placement of DBS and intracranial EEG electrodes — multi-trajectory cases take hours. **(partial robotic solution)**

Obstetric and Reproductive Care

- Operative vaginal delivery assistance — declining skill, high perineal trauma rates, shoulder dystocia management. **(no robotic solution exists)**
- Real-time fetal monitoring and intervention during labor — continuous CTG has poor specificity. **(no robotic solution exists)**
- Cervical cancer screening and Pap sample collection in low-resource settings — trained-personnel limited. **(no robotic solution exists)**
- Circumcision — high global volume; complication rates in adult MMC programs are non-trivial. **(no robotic solution exists)**

Bedside and Vascular Access Procedures

- Peripheral venous access and blood draws in difficult-access patients (pediatric, oncology, obese, elderly). **(no robotic solution exists)**
- Central venous catheter and arterial line placement. **(no robotic solution exists)**
- Hemodialysis fistula and graft cannulation — patients undergo this hundreds of times. **(no robotic solution exists)**
- Lumbar puncture and neuraxial anesthesia in difficult anatomy. **(no robotic solution exists)**
- Nasogastric and feeding tube placement — malposition causes serious harm. **(no robotic solution exists)**
- Thoracentesis, paracentesis, and lumbar drain placement. **(no robotic solution exists)**
- Endotracheal intubation in prehospital and difficult-airway settings. **(no robotic solution exists)**
- Ultrasound-guided regional anesthesia and chronic pain injections. **(no robotic solution exists)**

Endoscopy and Endovascular Procedures

- Distal neurovascular thrombectomy for stroke (M2/M3, A2, P2 occlusions). **(partial robotic solution)**
- Cardiac catheterization and PCI for stable coronary disease — robotic PCI exists but adoption is low. **(partial robotic solution)**
- Endoscopic retrograde cholangiopancreatography (ERCP) — post-procedure pancreatitis remains common. **(no robotic solution exists)**
- Routine upper GI endoscopy (EGD) — highest-volume endoscopy procedure. **(no robotic solution exists)**
- Colonoscopy completion and polypectomy in tortuous colons. **(partial robotic solution)**
- Bronchoscopic navigation *and* ablation of peripheral lung nodules — biopsy reach has improved; therapy delivery has not. **(partial robotic solution)**
- Percutaneous needle placement for tumor ablation in moving organs (liver, kidney, lung). **(no robotic solution exists)**

Critical Care and Emergency

- ICU patient repositioning and mobilization — pressure injury, VAP, and ICU-acquired weakness prevention. **(no robotic solution exists)**
- Mechanical ventilation titration and weaning — real-time adaptive control. **(no robotic solution exists)**
- Chest compression and resuscitation during transport with airway and drug delivery integration. **(partial robotic solution)**

Oncology, Radiotherapy, and Specimen Handling

- Brachytherapy seed and catheter placement (prostate, cervix, breast). **(no robotic solution exists)**
- External beam radiotherapy daily patient setup and immobilization. **(partial robotic solution)**
- Chemotherapy compounding and hazardous drug handling. **(partial robotic solution)**
- Intraoperative margin assessment in breast-conserving surgery. **(no robotic solution exists)**
- Bone marrow biopsy and aspiration. **(no robotic solution exists)**
- MRI-fusion prostate biopsy with per-lesion targeting. **(partial robotic solution)**
- Microbiology and pathology specimen processing in the laboratory. **(partial robotic solution)**

Imaging Acquisition

- Bedside transthoracic echocardiography by non-experts. **(no robotic solution exists)**
- Mammography and screening breast ultrasound positioning. **(no robotic solution exists)**

Dermatology, Dental, and Outpatient

- Dental implant placement and endodontic procedures. **(partial robotic solution)**
- Tympanostomy tube placement (currently requires general anesthesia in children). **(no robotic solution exists)**
- Skin biopsy and full-body lesion sampling for high-risk patients. **(no robotic solution exists)**
- Chronic wound debridement (diabetic ulcers, venous ulcers, pressure injuries). **(no robotic solution exists)**
- Burn depth assessment and skin graft harvesting/application. **(no robotic solution exists)**

Rehabilitation, Long-Term, and Home Care

- High-dose adaptive in-home neurorehabilitation after stroke or SCI. **(partial robotic solution)**
- Fall prevention, transfer, and mobility assistance for community-dwelling elders. **(no robotic solution exists)**

Targeted Drug Delivery

- Intracochlear and intravitreal sustained drug delivery. **(no robotic solution exists)**